

IBM BOB HACKATHON · NFSU · PROBLEM 11 · AI TRACK

NARCOBOB

Real-time narcotics hotspot intelligence, **reviewed by IBM Bob agents**

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github.com/vaibhavs-mishra/narcoBOB

SIMULATED FEED · real pipeline

THE PROBLEM

Seizure counts measure **police activity**,
not drug activity.

32.8

NDPS cases per lakh in Punjab

India's highest rate (NCRB 2021).
Intelligence reads these counts as
"where drugs are".

weeks

before a new route shows up

Overdose admissions sit in a
hospital silo; monthly statistics lag
the harm.

**harm >>
enf.**

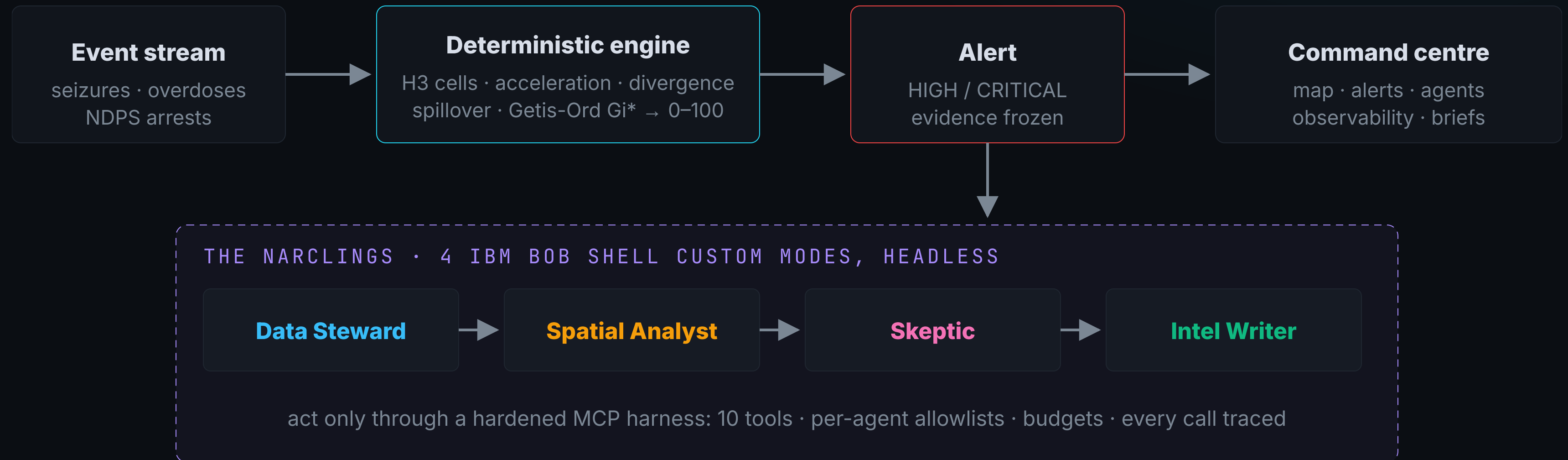
our signal

When overdoses rise and seizures
don't, a supply route is going
undetected. Score that, per
hexagon, every 2 s.

Predict **places, never people**, and recommend treatment alongside enforcement.

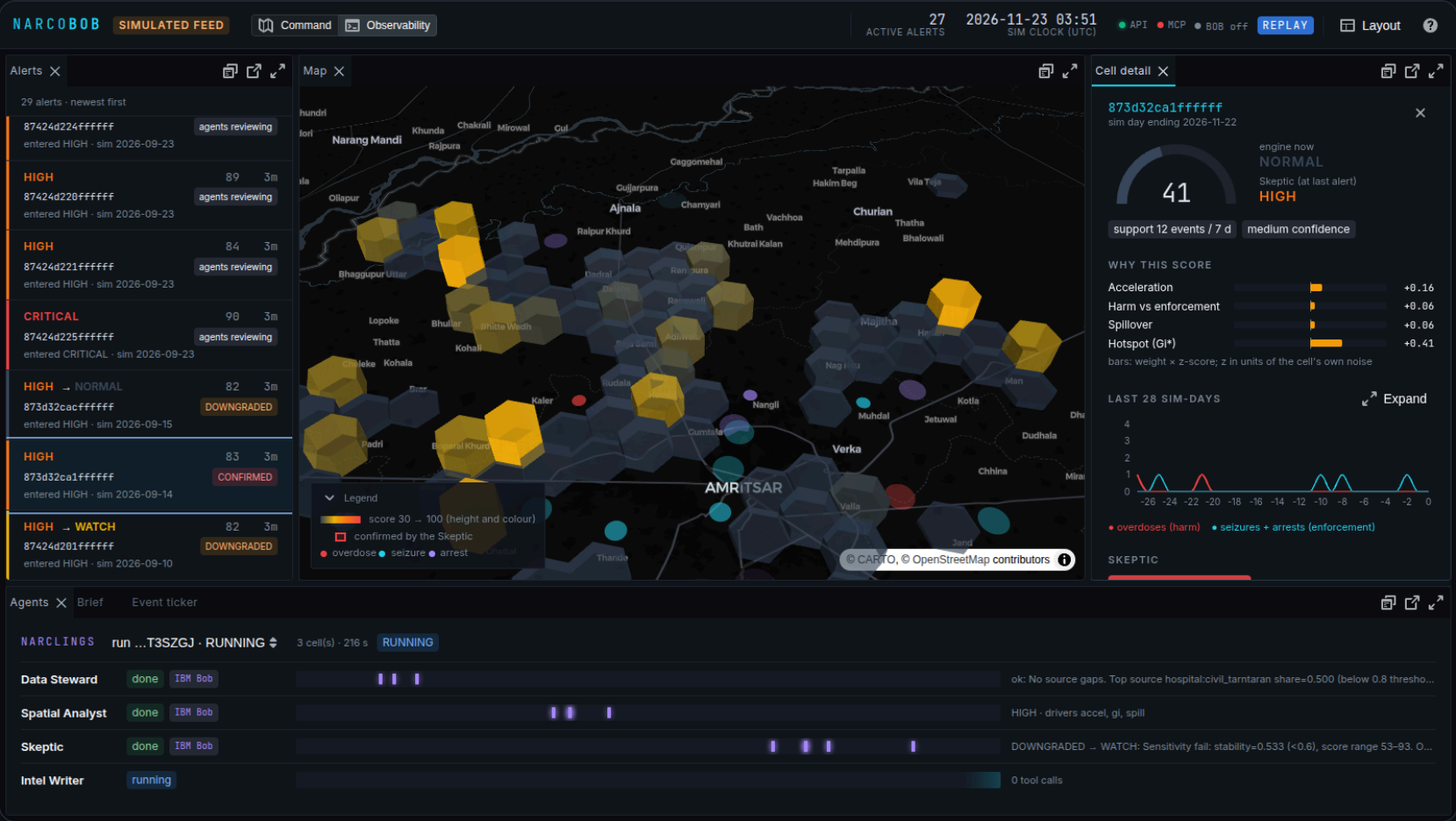
HOW IT WORKS

The engine does the maths. **Bob agents do the judgement.**

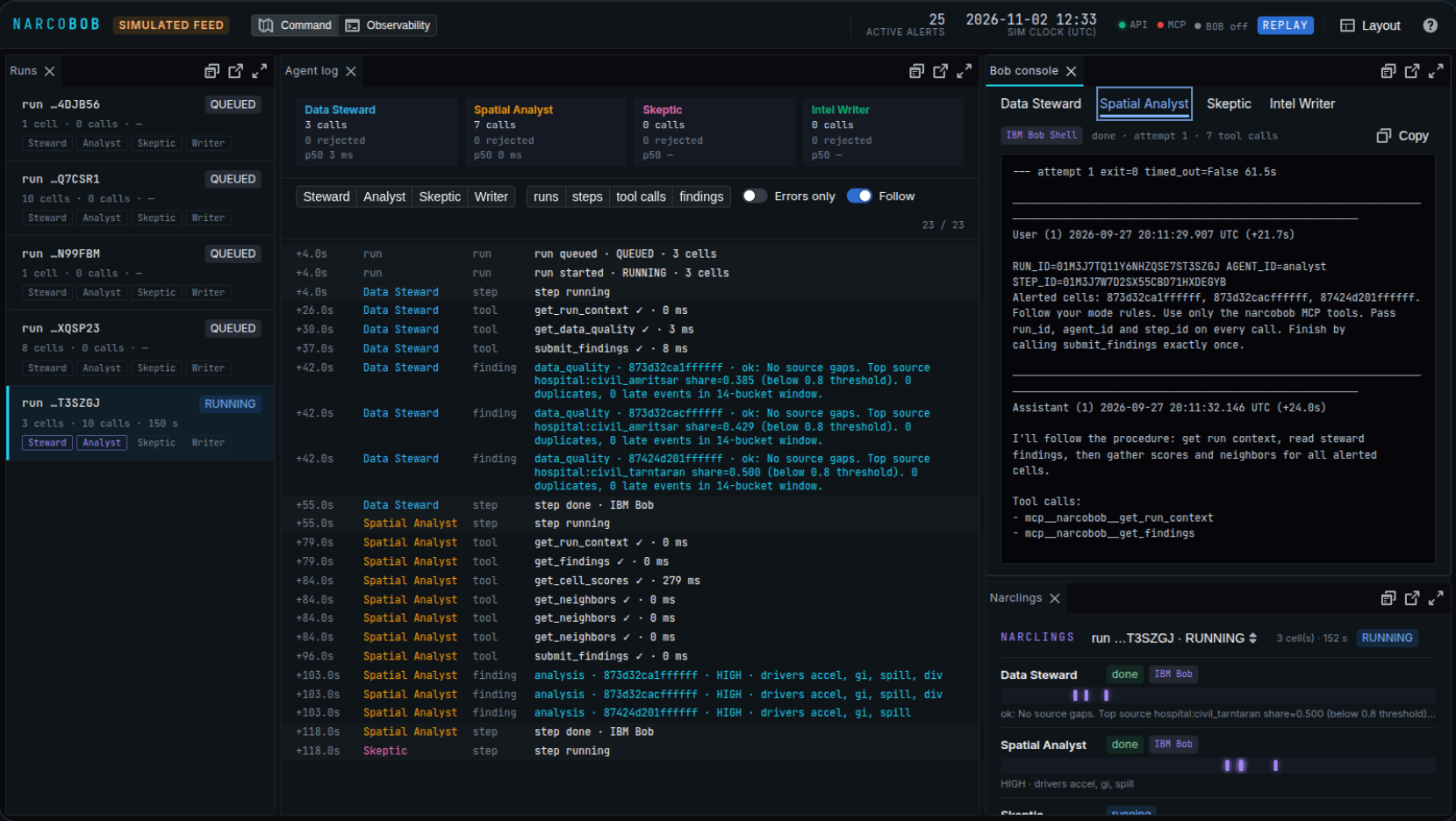


The Skeptic's verdict is the final severity. The Writer's brief cites only numbers returned by tools.

DEMO • A RECORDED REAL IBM BOB RUN



Alert → the Skeptic CONFIRMS the supply route; cell evidence and score drivers.



Observability: every step, MCP tool call and finding, plus each step's Bob console.

Video: demo/demo-video-link.txt • Run it: cd src && make setup && make replay

SIMULATED FEED

WHERE IBM BOB IS USED

Bob is **load-bearing**, not decorative.

The agents

4 Bob Shell **custom modes**, per-mode rules, shared `AGENTS.md`, `mcp.json`.

MCP tool group only: no edits, no shell.

The runtime

Every alert runs `bob run --mode` headless for each agent: Steward → Analyst → Skeptic → Writer. Checked, retried once, then fallback.

The guard-rails

Bob acts only through a 10-tool MCP harness: allowlists, budgets, tracing. Wrong tool → **FORBIDDEN_TOOL**.

The evidence

BOB light, *IBM Bob* step tags, Observability log + **Bob console**, briefs "written by IBM Bob agents".

Proof: `make bob-smoke` · `make bob-run` · `make dev` · `make replay` (a recorded real Bob session).

Every touchpoint, with links: `docs/ibm-bob-usage.md`

Finds the new route. Throws out the decoys.

PLANTED IN THE FEED	WHAT HAPPENED
New supply route: overdoses ×4 over 5 days, seizures flat	CONFIRMED · HIGH in 4 sim-days, CRITICAL in 5 · found on 12 / 12 seeds
A rural cell jumps from ~0 to 5 overdoses	Engine flags it · Skeptic DOWNGRADED (too few events)
A hospital stops reporting for 3 days	Skeptic DOWNGRADED (feed gap)
One-day police raid: seizures ×8, overdoses flat	No alert (a seizure ranking flags it)
Calm map, nothing planted	6.5 engine alerts / month → 3.8 after the Skeptic (raw rankings churn ~53)